Aagron School AI Operating System

Master Dataset Inventory | Autonomous Brain, Personal AI Teacher, Student Voice Intelligence, School Intelligence and shared AI governance

Capability	Categories
Autonomous Brain	**27**
Personal AI Teacher	**20**
Student Voice Intelligence	**18**
School Intelligence	**10**
Shared AI Training & Governance	**18**
TOTAL	**93**

This inventory contains 93 functional dataset categories. Categories are not separate AI models. Training data, live operational data, evaluations and confidential school records should be managed separately.

1. Aagron Autonomous Brain — 20 datasets

These datasets allow the system to understand, monitor, investigate, predict, and help manage school operations.

No.	Dataset name	Data to include
1	**School Master Dataset**	School type, location, curriculum, grades, sections, shifts, working hours, capacity, facilities
2	**School Organizational Structure**	Principal, coordinators, administrators, departments, reporting hierarchy, role responsibilities
3	**Staff Master Dataset**	Teacher and staff roles, qualifications, departments, subjects, availability, assigned responsibilities
4	**Student Enrollment & Capacity**	Class-wise enrollment, maximum seats, available seats, academic year, transfers
5	**Academic Calendar**	School days, holidays, examination dates, events, term dates, deadlines
6	**Timetable & Scheduling**	Class timetable, subject allocation, room allocation, teacher schedule, substitutions
7	**Staff Attendance & Leave**	Attendance, leave, absence reasons, approved leave, replacement availability
8	**Student Attendance Patterns**	Class-wise attendance, absenteeism trends, recurring patterns, authorized intervention records
9	**Classroom & Facility Availability**	Classroom capacity, room bookings, laboratory availability, facility utilization

No.	Dataset name	Data to include
10	**Asset & Equipment Management**	Asset inventory, condition, maintenance history, service life, warranties, breakdowns
11	**Maintenance & Infrastructure**	Repair complaints, fault categories, severity, maintenance schedules, resolution time
12	**Inventory & Procurement**	Stock levels, consumption, purchase requests, reorder points, vendor records, delivery history
13	**Vendor & Contractor Performance**	Contracts, service-level terms, delivery performance, complaints, costs
14	**Transportation Operations**	Vehicles, routes, schedules, utilization, maintenance, approved driver and safety records
15	**Hostel & Campus Services**	Hostel capacity, room allocation, meal services, sanitation, campus service issues
16	**Finance & Cost Intelligence**	Department budgets, spending, fee collection, operating expenses, utility costs, cost-saving opportunities
17	**Admission & Expansion Intelligence**	Historical admissions, enquiries, conversion, waiting lists, dropout trends, capacity and demand projections
18	**Safety, Incidents & Compliance**	Safety reports, emergency procedures, incident classifications, mandatory inspections, regulatory obligations
19	**School Resource Network**	Underused or unavailable resources, approved partner schools, borrowing permissions, shared facilities, transport and logistics
20	**Historical Operational Problems & Outcomes**	Previous school problems, underlying causes, attempted solutions, assignments, timelines, verified results, recurrence

Autonomous Brain — AI reasoning and execution datasets

In addition to the 20 operational source datasets, build these seven specialized datasets to train and evaluate its decision-making behavior.

No.	Dataset	What the AI must learn
21	**Problem Detection**	Identify operational failures, unusual patterns, missing resources, and emerging risks
22	**Root-Cause Investigation**	Distinguish symptoms from actual causes using available evidence
23	**Solution Generation**	Produce practical solutions based on resources, constraints, policies, and historical outcomes
24	**Scenario Simulation**	Compare feasible options, likely consequences, uncertainty, and failure conditions
25	**Prioritization & Risk**	Classify urgency, impact, student safety, dependencies, and escalation requirements
26	**Assignment & Follow-up**	Identify authorized staff, assign tasks, track deadlines, follow up, and escalate delays

No.	Dataset	What the AI must learn
27	**Verification & Prevention**	Confirm the work is complete, test whether the problem is resolved, and identify measures to prevent recurrence

Autonomous Brain total: 27 dataset categories.

Its training examples should cover common and difficult situations: sudden teacher absence, classroom overcrowding, timetable clashes, broken equipment, laboratory shortages, transport disruptions, safety issues, unexpected increases in expenditure, vendor failures, insufficient staffing, and recurring unresolved complaints.

A critical distinction: the model can learn how to investigate and recommend, but staff authorizations, safety requirements, spending limits, and execution permissions must be enforced through reliable software rules—not left to AI judgment.

2. Aagron Personal AI Teacher — 20 datasets

These datasets power the Personal AI Teacher inside the existing Student Portal.

No.	Dataset name	Data to include
1	**Curriculum & Board**	CBSE, ICSE, state boards, approved curricula, academic years, grade-specific requirements
2	**Class, Subject & Chapter**	Grade, subject, chapter, unit, topic, subtopic, prerequisite relationships
3	**Learning Objectives**	Required knowledge, expected skills, competencies, mastery criteria
4	**Textbook & Study Material**	Authorized textbook content, explanations, diagrams, examples, references
5	**Question Bank**	MCQs, short answers, long answers, numerical problems, case studies
6	**Correct Answers & Rubrics**	Verified answers, marking schemes, acceptable alternative answers, partial-credit rules
7	**Worked Solutions**	Step-by-step solutions, calculations, reasoning, common solution methods
8	**Student Academic History**	Subject marks, chapter performance, prior assessments, improvement trends
9	**Assessment & Answer Analysis**	Questions attempted, student answers, evaluation results, teacher corrections
10	**Learning Gap Identification**	Missing concepts, incorrect assumptions, weak topics, unmet prerequisites
11	**Misconception & Error Patterns**	Common mistakes, calculation errors, conceptual confusion, recurring misconceptions
12	**Explanation & Teaching Methods**	Age-appropriate explanations, examples, analogies, alternative teaching approaches
13	**Adaptive Learning Paths**	Topic sequence, prerequisite remediation, personalized revision steps

No.	Dataset name	Data to include
14	**Practice & Reinforcement**	Practice exercises, difficulty levels, spaced revision, hints, retry rules
15	**Learning Conversations**	Student questions, approved AI responses, clarification requests, follow-up explanations
16	**Multilingual Education**	English, Hindi, regional-language explanations, subject terminology, translation quality
17	**Student Engagement Signals**	Practice frequency, completion, requests for help, time spent, voluntary feedback
18	**Learning Accessibility**	Accessible content formats, reading-level adjustments, approved learning accommodations
19	**Mastery & Improvement Verification**	Independent reassessment, error reduction, retention over time, mastery evidence
20	**Pedagogical Safety & Accuracy**	Hallucination detection, incorrect explanations, age-appropriate boundaries, uncertainty and escalation

Personal AI Teacher total: 20 dataset categories.

The most important training chain is:

Student answer + curriculum topic + verified assessment result

Identify the misconception and its likely learning prerequisite

Generate an age-appropriate explanation and targeted practice

Deliver through the student's Personal AI Teacher

Reassess independently, verify improvement, and adapt

Train the teacher to explain concepts correctly rather than simply supply answers. Include deliberately challenging examples involving ambiguous questions, multiple valid solutions, incorrect answer keys, and students who do not improve after repeated practice.

3. Aagron Student Voice Intelligence — 18 datasets

These datasets support two core functions: school-to-student communication through messages, voice recordings, and forms; and confidential student concerns, replies, solutions, and staff follow-up.

No.	Dataset name	Data to include
1	**Student Communication Profiles**	Class, section, communication language, authorized contact and delivery preferences
2	**School Messages**	Announcements, teacher messages, notices, student-specific communications
3	**Voice Recording & Transcription**	Consented speech samples, transcripts, language, transcription errors, noisy audio
4	**Multilingual Voice Understanding**	Hindi, English, regional languages, code-switching, accents, classroom terminology

No.	Dataset name	Data to include
5	**Student Forms & Surveys**	Questions, response types, form templates, audience selection, response deadlines
6	**Message–Reply Linking**	Original message or form, related student response, thread history, delivery status
7	**Student Concern Categories**	Academic concerns, bullying, harassment, safety, discrimination, unfair treatment, school facilities
8	**Academic Distress & Feedback**	Difficult subjects, low-mark concerns, exam anxiety, perceived academic difficulties
9	**Teacher–Student Relationship Issues**	Communication difficulties, classroom treatment, unmet requests for help, disputes
10	**Peer & Classroom Issues**	Bullying, exclusion, classroom disruptions, peer conflicts
11	**Student Wellbeing Signals**	Self-reported distress, requests for support, uncertainty indicators, need for human review
12	**Sentiment, Intent & Context**	Complaint intent, requests, urgency, uncertainty, context-dependent meanings
13	**Severity & Safeguarding**	Immediate safety risks, protection requirements, confidential escalation criteria
14	**Concern Investigation**	Relevant evidence, missing information, contradictory reports, safe follow-up questions
15	**Solution Recommendations**	Appropriate support options, school policies, historical outcomes, practical next steps
16	**Authorized Staff Routing**	Welfare staff responsibilities, permissions, availability, escalation channels
17	**Case Follow-up & Resolution**	Assigned personnel, action taken, response time, student feedback, verified resolution
18	**Privacy, Consent & Confidentiality**	Access permissions, consent status, retention, data minimization, restricted case sharing

Student Voice Intelligence total: 18 dataset categories.

Aagron should train on ordinary communication as well as difficult cases: unclear audio, incomplete disclosures, students withdrawing complaints, false or contradictory information, repeated unresolved concerns, and situations where immediate safeguarding action is needed.

For serious student safety concerns, AI must not independently investigate or conclude that a case is resolved. It should route information to an appropriately authorized human safeguarding professional using established procedures.

4. Aagron School Intelligence — 10 datasets

These datasets connect information across the school to recognize broader trends without exposing individual students' confidential information unnecessarily.

No.	Dataset name	Data to include
1	**School-Wide Academic Trends**	Aggregated learning gaps, subject-level achievement, class-level improvement
2	**Classroom Environment Patterns**	Aggregated feedback, classroom conditions, teaching-resource availability
3	**School-Wide Student Feedback**	Privacy-protected themes from surveys, voice feedback, and complaints
4	**Cross-Department Problems**	Problems involving multiple departments, shared dependencies, recurring failures
5	**School Climate & Welfare Trends**	Privacy-protected concern trends, resolution delays, recurring support needs
6	**Staff & Resource Utilization**	Timetable coverage, workloads, facilities, equipment and allocation patterns
7	**Intervention Effectiveness**	Which educational and operational interventions improved verified results
8	**Improvement Opportunity Identification**	Recurring bottlenecks, preventable costs, unused resources, unmet academic needs
9	**School Research & Recommendations**	Evidence summaries, possible interventions, resource requirements, uncertainty
10	**Long-Term School Improvement**	Monthly and annual trends, intervention follow-up, effectiveness, recurrence monitoring

School Intelligence total: 10 dataset categories.

This is the layer that can recognize, for example, that many students in different sections have the same mathematics misconception or that the same operational issue repeatedly disrupts classes.

Such findings must use appropriate aggregation and privacy thresholds. A school-wide dashboard should not reveal an individual student's sensitive complaint through a small group or identifiable pattern.

5. Common AI Training, Evaluation & Governance — 18 datasets

These are shared across all four capabilities. Without them, the system may have school data but still lack reliable training examples, quality controls, or evidence that it works.

No.	Dataset name	Purpose
1	**AI Instruction Dataset**	Teach expected response format, permitted tasks, workflow discipline
2	**Tool-Use & Workflow Dataset**	Teach when to query school systems, calculate, create tasks, or request human action
3	**Input-to-Outcome Examples**	Connect a school situation, evidence, decision and verified result
4	**Multi-Step Reasoning Examples**	Test investigation, planning, dependency handling, and solution comparison

No.	Dataset name	Purpose
5	**Domain Policies & Rules**	School rules, authorization limits, academic policies, safeguarding protocols
6	**Human Corrections & Expert Labels**	Teacher corrections, administrator review, welfare expert assessments
7	**Positive & Negative Examples**	Successful actions, failed actions, unsafe outputs, inaccurate conclusions
8	**Rare-Event & Edge Cases**	Unusual combinations, emergencies, missing records, complex scheduling conflicts
9	**Synthetic Scenario Dataset**	Carefully generated fictional school events for training and testing
10	**Ground-Truth Evaluation Set**	Verified expected results to measure system performance objectively
11	**Adversarial & Security Tests**	Prompt injection, impersonation, unauthorized requests, data leaks
12	**Bias & Fairness Evaluation**	Performance across grades, languages, accessibility needs and school contexts
13	**Confidence & Uncertainty Dataset**	Insufficient evidence, conflicting information, abstention, escalation
14	**Privacy & Consent Evaluation**	Sensitive-data handling, consent enforcement, role-based disclosure
15	**Tool Failure & Recovery**	API outages, duplicate actions, failed notifications, stale or incomplete data
16	**Human Override & Audit Trail**	Reversed decisions, corrections, accountable actions, audit evidence
17	**Model Performance Monitoring**	Accuracy, false alarms, unsafe responses, latency, cost, model version
18	**Continuous Learning Feedback**	Verified new outcomes, distribution changes, approved model updates

Implementation notes

Maintain a documented source, schema, consent or other lawful processing basis, allowed uses, retention period, validation rules, verified labels, and independent evaluation set for each dataset. Keep school-specific live data permission-controlled, and do not assume that confidential student data may be used for model training.

Recommended build order: foundation and integrations; Autonomous Brain; Personal AI Teacher; Student Voice Intelligence; School Intelligence; evaluation and controlled monthly improvement.